

Map of Utility-Scale Solar Installations

- Intended Use, Limitations, and Suggested Workflow

Kangjoon Cho, Curtis Woodcock

<https://kangjoon.projects.earthengine.app/view/bu-celtmass-eea---solar-mapping>

Purpose

This map was developed as part of an effort to estimate statewide area and uncertainty of utility-scale solar installations and associated land-cover change across Massachusetts using Landsat time-series imagery (2005-2024). Find more details in the published paper below:

<https://www.sciencedirect.com/science/article/pii/S2666017225001439>

Map contents

The map contains land-cover change classes that can support different aspects of the Massachusetts solar inventory.

Class	Description	Potential application for screening
Solar with deforestation	Utility-scale solar developments established following forest clearing	Identify additional ground-mounted solar facilities that may be missing from the inventory / Identify what type of land-cover change is associated
Solar w/o deforestation	Utility-scale solar developments established on previously non-forested land.	Identify additional ground-mounted solar facilities that may be missing from the inventory
Deforestation nearby solar	Forest clearing adjacent to mapped solar developments	Review whether existing polygons fully capture the overall development footprint and associated land-cover change
Potential solar	Locations with spectral characteristics similar to solar developments but missing in the classification results (For accuracy assessment purpose)	Identify additional ground-mounted or building-mounted solar facilities that may be missing from the inventory
Buffer of solar	A buffer (5 pixels) surrounding mapped solar developments	-
Not Solar Change	Land-cover changes not associated with solar development	-

Suggested applications

- Identify utility-scale solar facilities that may be missing from the current inventory
- Prioritize polygons for review and digitization
- Identify locations where existing polygons may not fully represent the developed area or associated land-cover changes

Detection capability

The mapping approach was designed for utility-scale developments (>1 MWdc)

- Facilities around 150 kWdc typically occupy only 3-4 Landsat pixels, making reliable detection difficult
- Objects smaller than 7 Landsat pixels were intentionally removed during the mapping process.
- Ground-mounted solar facilities larger than approximately 300 kWdc may be detected, although detection becomes substantially more reliable for facilities larger than 500 kWdc.

Key limitations

- Small facilities (approximately 150-300 kWdc) are frequently below the reliable detection threshold.
- Polygon boundaries represent development footprints and may require refinement using high-resolution imagery. (Area estimation is the main purpose of this mapping rather than getting precise footprints)
- Some commission errors are expected.

For the Mass EEA database

This map can substantially reduce the effort required to locate missing utility-scale facilities. When a solar development appears on this map but is absent from the existing inventory, it can often be digitized directly from high-resolution imagery instead of relying on addresses or permitting records.

Likewise, differences between the mapped area and collected polygons can help identify sites where additional land-cover changes or facility boundaries should be reviewed.

Suggested workflow and issues

1. Use the statewide solar map as an independent source to identify utility-scale facilities (or facilities with >300-500 kWdc) that may be missing from the inventory.
2. Compare mapped solar pixels with existing polygons to identify sites where boundaries or associated land-cover changes may require further review.
3. Prioritize manual inspection of locations highlighted by the map to improve the efficiency of inventory updates and quality assurance.
4. We need alternative approaches to identify facilities in the 150-300 kWdc range, which are generally below the reliable detection capability of this Landsat-based mapping approach.